Base service station (BSS) colour: This system identifies adjacent networks and marks them. Signal filtering is made easy and network conflicts are minimised. In enterprise systems, it allows segregated networks to cover the same physical area without losses due to interference.

Target wake time (TWT): This feature is unique to WiFi 6. It allows each device to independently negotiate wake time for transmission and reception. This helps to increase total sleep time and maximise battery life. TWT enables many additional networking options, especially for IoT devices.

***Improved security*:** All WiFi 6 devices will need to include Wi-Fi Protected Access 3 (WPA3). This will lead to encryption of unauthenticated traffic, robust password protection against brute-force dictionary attacks, and superior data reliability for sensitive information with 192-bit encryption.

Beamforming: With eight support antennas, beamforming helps to improve data rates, and the range is extended by directing signals towards specific clients at once. It offers a backup for rapidly moving devices that may face issues with MU-MIMO. Beamforming also helps to control transmissions from antennas that cause signals to interfere on purpose. The signal can then be redirected to a new direction.

Devices that support WiFi 6

Until recently, WiFi 5 was the standard used for routers, repeaters, mesh networks and many WiFi clients. WiFi 6 was launched in 2019. There will be some compatibility issues for the earlier devices that supported WiFi 5 — they will be able to utilise the WiFi 6 network but not be able to receive support for the same.

WiFi 6 routers are backward-compatible, and it is better to make sure that the network is ready for that.

WiFi 6 enables lower battery consumption, making it a great choice for any environment, including the Internet of Things (IoT). It reduces unnecessary data activity, and tells devices when to put their data to sleep and when to be active. As a result, unnecessary data activity is reduced, and performance and battery life are maximised.

The Samsung Galaxy Note 10 and Ruckus R750 were the world's first smartphone and access point certified to support Wi-Fi 6, with the latest generation of the Apple iPhone following suit. The Wi-Fi Alliance has set up its certification programme, and new wireless products hitting the market are expected to start applying for compliance certification. The devices listed below are already WiFi 6 enabled:

- iPhone 11 and after
- Samsung Galaxy S10, S20, Note 10, and Note 20
- Apple computers with M1 processors
- Smart TVs

> To take advantage of the improvements in the 802.11ax standard fully, both hardware and software functionalities have to be built on this WiFi technology.

Hardware testing

To unlock the full potential of the latest devices, a WiFi 6 router is needed to run the network. This was an expensive affair a few years ago, but now we have a number of options even for mesh systems, gaming routers, range extenders, and more. The best purchase can be made only when hands-on testing is done. Beating all its competitors, the current king in terms of speed for WiFi 6 routers is TP-Link Archer AX6000. This router was able to transmit data wirelessly at a rate of 1523 Mbps up to a distance of 1.5 metres (5 feet).

One important thing to remember here is that these routers do not magically increase speeds. The theoretical maximum of achieving 9.6 Gbps is unlikely. This high theoretical speed can be split up across a whole network of devices.

WiFi 6 emphasises quality connectivity in areas where connected devices are densely populated. It does not increase the speed of each device exponentially but ensures these operate at an optimum level.

Only the combination of a faster plan from the Internet service providers (ISPs) along with the WiFi 6 router, can fulfil its true potential. The real challenge is for the ISPs, as they need new fibre rollouts to capitalise on this next-gen technology. An important question is: when faster ISP speeds come, will the existing hardware become redundant?

Applications of WiFi 6

Large public venues (LPVs): Stadiums and convention centres are a few of the common areas where thousands of devices connect to WiFi at the same time. WiFi 6 can help to improve attendee experiences, increase customer interactions, and create value-added services like viewing instant replays or ordering food from one's seat at an event. WiFi 6 allows LPV owners to create new business opportunities.

Transport hubs: Public transport stations are also an area where people attempt to connect to the network simultaneously. OFDMA and BSS colouring in WiFi 6 provide the necessary tools needed to overcome this challenge.

IoT and smart city deployments: Power efficiencies in WiFi 6 enable IoT devices to go into sleep mode and turn on their transmitters at predefined intervals to prolong field time without much maintenance.

Education: Libraries, auditoriums, and lecture halls at college and university campuses have the highest density of WiFi users during the day, and almost no one at night. WiFi 6 is a perfect choice in this situation.